

## Problem 1 — Linear Maps, RREF, Column Relations

**T1: Find RREF of a given matrix.**

Eliminate below pivots  $\rightarrow$  above  $\rightarrow$  scale to 1. # pivots = rank. Free columns = cols without a pivot.

**T2: Geometric map  $\rightarrow$  find its matrix.**

Track each  $e_i$ ; column  $i$  of matrix =  $Ae_i$ .

- Projection onto  $n^\perp$ :  $P = I - \frac{nn^\top}{n^\top n}$
- Reflection across  $n^\perp$ :  $H = I - \frac{2nn^\top}{n^\top n}$ ;  $H^2 = I$ ,  $H^{-1} = H$
- 2D rotation by  $\theta$ :  $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$
- Shearing, scaling: write directly from geometric action on  $e_i$ .

**T3: Real-life setup (chicken-rabbit style)  $\rightarrow$  matrix.**

Unknowns = variables, constraints = equations. Write  $Ax = b$ , row-reduce.

**T4:  $A = [u \ v \ w]$  described by column relations.**

Translate relations into column equations. E.g.  $w = u + v \Rightarrow$  col 3 of RREF has  $(\dots, 1, \dots, 1)^\top$  pattern. RREF directly encodes all column dependencies.

**T5: Partial matrix + RREF  $\rightarrow$  recover full  $A$ .**

RREF tells you: “free col = linear combo of preceding pivot cols.” E.g. if RREF col  $j = (c_1, c_2, c_3, 0)^\top$ , then  $a_j = c_1 a_{\pi_1} + c_2 a_{\pi_2} + c_3 a_{\pi_3}$  (where  $\pi_i$  are pivot col indices). Use known entries to anchor pivot columns, reconstruct the rest.

**T6: Row/column operations to achieve a result.**

Elementary row-op “add  $c \times \text{row}_j$  to  $\text{row}_i$ ”: matrix  $E =$  identity with  $E_{ij} = c$ . Col ops: multiply on right. Chain left-to-right for rows, right-to-left for cols. **Trick:** to clear all non-pivot entries from RREF,

the column-op matrix  $C$  is unit upper triangular; each non-pivot column of  $C$  is  $-(\text{RREF column without pivot entry})$ .

**T7: Solve  $Ax = b$ .**

Row-reduce  $[A \mid b]$ . Check consistency (no pivot in augmented col). General soln:  $x = x_p + x_h$  where  $x_p =$  any particular soln,  $x_h \in \ker(A)$  (from free vars).

**T8: Rank questions.**

$\text{rank}(A) = \text{rank}(A^\top)$ ;  $\text{rank}(AB) \leq \min(\text{rank} A, \text{rank} B)$ ;  $\text{rank}(A^\top A) = \text{rank}(A)$ . Invertible  $C$ :  $\text{rank}(AC) = \text{rank}(A)$ . Key: if  $C - I$  is nilpotent ( $\exists k, (C - I)^k = 0$ ), then  $(C - I)^{k+j} = 0$  for all  $j \geq 0$ .

## Problem 2 — Inverses, LDU, Matrix Algebra

**T1: Find inverse or LDU decomposition.**

Forward elim: record multipliers in  $L$  (unit lower), diagonal pivots in  $D$ , unit upper in  $U$ . **LDU exists  $\iff$  every leading principal minor is nonzero.** Inverse via row-reduction:  $[A \mid I] \xrightarrow{\text{row ops}} [I \mid A^{-1}]$ .

**T2: Which products are legal?**

$(m \times n)(n \times p) \rightarrow m \times p$ . Inner dims must match. Check left-to-right.

**T3: Inverse/LDU of a product.**

$(AB)^{-1} = B^{-1}A^{-1}$ .  $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$ . For LDU of  $A^\top A$ : compute  $A^\top A$  first, then decompose. If  $A$  has full col rank,  $A^\top A$  is invertible.

**T4: Given  $A^{-1}$ , find  $A$ ; or given partial info about inverse.**

$(A^{-1})^{-1} = A$ . Reverse row ops. Or: last row of  $A^{-1}$  must be  $\perp$  to all cols of  $A$  except the last; use cross-product / dot conditions.

**T5: Partial matrix, test when LDU fails.**

Find parameter value(s) making a leading minor zero. At those values, no LDU exists.

**T6: Unorthodox inverse methods.**

**Col-op method:** column ops on  $\begin{pmatrix} A \\ I \end{pmatrix}$ ; when top block  $\rightarrow I$ , bottom  $\rightarrow A^{-1}$ .

**Sherman-Morrison (rank-1 update):**  $(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u}$

**Pseudo-inverse** ( $A$  full col rank):  $B = (A^\top A)^{-1}A^\top$ . Proof:  $ABA = A((A^\top A)^{-1}A^\top A) = A$ . Also  $AB = A(A^\top A)^{-1}A^\top$  is orthogonal projection  $\Rightarrow (AB)^2 = AB$ .

**Polynomial of a projection:**  $(AB)^k = AB$  for  $k \geq 1$ . So  $p(AB) = p(1) \cdot AB + p(0) \cdot (I - AB)$ . E.g.  $p(x) = x^4 - 3x^2 + 2x + 1 \Rightarrow p(AB) = (1 - 3 + 2 + 1)AB + 1 \cdot (I - AB) = I$ .

## Problem 3 — Block Matrix Operations

**T1: Block ops, block LDU, block multiplication.**

Treat blocks like scalars — **but respect order**. If  $R$  invertible:  $\begin{pmatrix} R & A \\ A^\top & O \end{pmatrix} = LDU$  with  $L = \begin{pmatrix} I & 0 \\ A^\top R^{-1} & I \end{pmatrix}$ ,  $D = \begin{pmatrix} R & 0 \\ 0 & S \end{pmatrix}$ ,  $U = \begin{pmatrix} I & R^{-1}A \\ 0 & I \end{pmatrix}$  where  $S = -A^\top R^{-1}A$  (Schur complement). Block row-swap:  $P = \begin{pmatrix} O_{r \times s} & I_{r \times r} \\ I_{s \times s} & O_{s \times r} \end{pmatrix}$ . List sizes of every block. To kill the  $(1, 2)$  block  $B$  using block  $(2, 2) = D$ : subtract  $\text{row}_1 - BD^{-1} \times \text{row}_2$ .

**T2: Spot blocks in a big matrix, then apply block methods.**

Look for: zero blocks, repeated submatrices, rectangular structure. Rank of  $M \leq$  sum of ranks of diagonal blocks after block-triangularization. For  $M = \begin{pmatrix} A & AB \\ BA & B \end{pmatrix}$  ( $A$  is  $3 \times 2$ ,  $B$  is  $2 \times 3$ ): rank  $\leq 4$ . Row op kills upper-right:  $\begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} M = \begin{pmatrix} A - BA & 0 \\ BA & B \end{pmatrix}$  — but simpler: note  $M$  has col span in the 5-dim space of cols of  $\begin{pmatrix} A \\ BA \end{pmatrix}$  and  $\begin{pmatrix} AB \\ B \end{pmatrix}$ .

**T3: Find block  $X$  s.t.  $MX = N$  (reverse-engineer).**

Strategy: factor  $N = M \cdot X$  by decomposing  $N$  into terms each matching block rows of  $M$ , or augment  $[M \mid N]$  and block-eliminate to get  $[I \mid X]$  (or  $[R \mid X']$ , then read off).

## Problem 4 — Fundamental Subspaces, Projections

**T1: Find the four fundamental subspaces of  $A$ .**

Compute  $\text{RREF}(A)$ :

- **Ran**( $A$ ) (col space): take *original* cols at pivot positions.
- **Ker**( $A$ ) (null space): for each free var  $x_i = 1$  (others = 0), back-substitute  $\rightarrow$  basis vector.
- **Ran**( $A^\top$ ) (row space): nonzero rows of  $\text{RREF}$  are a basis.  $\text{Dim} = \text{rank}$ .
- **Ker**( $A^\top$ ) (left null): row-reduce  $A^\top$ , or use  $\text{dim} = \# \text{rows} - \text{rank}$ .

**Rank-nullity:**  $\text{rank}(A) + \text{nullity}(A) = \# \text{cols}$ .

**T2: Subspaces via constraint equations  $\rightarrow$  basis,  $V \cap W, V + W$ .**

Encode each subspace as solution set of  $Cx = 0$ .

- $V \cap W$ : stack both systems  $\rightarrow$  find null space.
- $V + W$ : concatenate basis vectors of  $V$  and  $W$ , reduce to find basis.
- $\text{dim}(V + W) = \text{dim } V + \text{dim } W - \text{dim}(V \cap W)$ .

If a kernel  $B$  has  $\text{ker}(B) = V + W$ : use rank to determine rank of block matrices involving  $B$ .

**T3: Projection matrices.**

- Onto  $\text{Ran}(A)$  (full col rank):  $P = A(A^\top A)^{-1}A^\top$
- Onto  $\text{Ker}(A^\top)$ :  $I - P$  (complement projection)
- Onto  $\text{span}(w)$ :  $P = ww^\top / (w^\top w)$
- Projection  $P$  satisfies:  $P^2 = P, P^\top = P, I - P$  also a projection.

**Tip:** If  $A$  is  $m \times n$  with rank  $r$ , projection onto  $\text{Ker}(A^\top)$  is  $I_m - P$  and has rank  $m - r$ .

**Finding  $Ax = b$  solutions + projection:**

- If  $b \notin \text{Ran}(A)$ : no solution; best approx via  $\hat{x} = (A^\top A)^{-1}A^\top b$  (least squares).
- If  $b \in \text{Ran}(A)$ : complete solution  $x = x_p +$  (null space of  $A$ ).

## Problem 5 — Abstract Spaces, Bases & Change of Coords

**T1: Function / differential-equation spaces.**

If basis is  $\mathcal{B} = (f_1, f_2)$ , represent  $f = c_1 f_1 + c_2 f_2$  as vector  $(c_1, c_2)^\top$ . Apply linear map (e.g.  $D = d/dx$ ) to each basis element, express in target basis  $\rightarrow$  columns.

- $D(\sin 2x) = 2 \cos 2x, D(\cos 2x) = -2 \sin 2x$
- Matrix of  $D$  under  $\mathcal{B} = (\sin 2x, \cos 2x)$ :  $\begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$
- Verify:  $f + 4f'' = 0 \Rightarrow$  solution space spanned by  $(\sin 2x, \cos 2x)$ .

**T2: Matrix spaces  $V = M_2(\mathbb{R})$ , map  $T(X) = AX + XA$ .**

Standard basis  $\mathcal{B} = (E_{11}, E_{12}, E_{21}, E_{22})$  where  $E_{ij} = e_i e_j^\top$ . Apply  $T$  to each  $E_{ij}$ : compute  $AE_{ij} + E_{ij}A$ , express as vector in  $\mathcal{B} \rightarrow$  column of matrix.

**T3:  $\mathbb{R}^n$  under an exotic basis  $\mathcal{B} = (b_1, \dots, b_n)$ .**  
 $P = [b_1 \mid \dots \mid b_n]$  (change-of-basis matrix,  $\text{std} \rightarrow \mathcal{B}$ ). Coordinate vector:  $[x]_{\mathcal{B}} = P^{-1}x$ . Map  $T$  under  $\mathcal{B}$ :  $[T]_{\mathcal{B}} = P^{-1}T_{\text{std}}P$ .

**T4: Change-of-coordinate matrices.**

$P_{\mathcal{C} \leftarrow \mathcal{B}}$ : express each  $\mathcal{B}$ -vector in  $\mathcal{C}$ -coords, stack as columns.  $P_{\mathcal{B} \leftarrow \mathcal{C}}: (P_{\mathcal{C} \leftarrow \mathcal{B}})^{-1}$ . Relation between matrix reps:  $T_{\mathcal{B} \leftarrow \mathcal{B}} = P_{\mathcal{B} \leftarrow \mathcal{C}} T_{\mathcal{C} \leftarrow \mathcal{B}}$ . More generally:  $T_{\mathcal{C}' \leftarrow \mathcal{B}'} = P_{\mathcal{C}' \leftarrow \mathcal{C}} T_{\mathcal{C} \leftarrow \mathcal{B}} P_{\mathcal{B} \leftarrow \mathcal{B}'}$ .

**T5: Linear map between different spaces.**

Matrix of  $T: V \rightarrow W$  w.r.t.  $\mathcal{B}_V, \mathcal{B}_W$ : apply  $T$  to each  $\mathcal{B}_V$  vector, express in  $\mathcal{B}_W$ , stack as columns.

**T6: Trig basis change via angle-addition formulas.**

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \quad \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Under  $\mathcal{B} = (\sin 2x, \cos 2x)$ :  $\sin(2x + \phi) \leftrightarrow \begin{pmatrix} \cos \phi \\ \sin \phi \end{pmatrix}$ ;  $\cos(2x + \phi) \leftrightarrow \begin{pmatrix} -\sin \phi \\ \cos \phi \end{pmatrix}$ .

Under  $\mathcal{C} = (\sin(2x + \phi), \cos(2x + \phi))$ : change-of-basis matrix from  $\mathcal{B}$  to  $\mathcal{C}$  involves  $\pm \sin \phi, \cos \phi$ .

**Linearity check (always same proof):**

$T(X + Y) = T(X) + T(Y)$  and  $T(kX) = kT(X)$  — hold by distributivity of the underlying operations.

**Nilpotency and powers of  $T$ :**

If  $A^2 = O$ , then  $T(X) = AX + XA$  satisfies  $T^3 = O$ :  $T^3(X) = A^3X + 3A^2XA + 3AXA^2 + XA^3 = 0$  (all terms have  $A^2$  factor). More generally if  $A^k = O$ :  $T^{2k-1} = O$  (binomial expansion of  $(L_A + R_A)^{2k-1}$ ).

## Problem — Universal Quick Reference

**Invertibility:**

- Invertible  $\iff$  full rank  $\iff \text{RREF} = I$ , trivial kernel
- LDU: all leading minors  $\neq 0$
- $(AB)^{-1} = B^{-1}A^{-1}$ ;  $(AB)^\top = B^\top A^\top$

**Projections:**

- $P^2 = P$ ;  $(I - P)^2 = I - P$
- $p(P)$ : use  $P^k = P$  for  $k \geq 1$ ,  $(I - P)^k = I - P$
- $(AB)^k = AB$  when  $ABA = A$

**Four Subspaces:**

$$\begin{aligned} \mathbb{R}^n &= \text{Ran}(A^\top) \oplus \text{ker}(A) \\ \mathbb{R}^m &= \text{Ran}(A) \oplus \text{ker}(A^\top) \end{aligned}$$

**Nilpotent matrix**

$$N^k = O: (I + N)^{-1} = I - N + N^2 - \dots + (-1)^{k-1} N^{k-1}$$

**Block rank:**

$$\text{rank} \begin{pmatrix} A & AB \\ BA & B \end{pmatrix} \leq \text{rank}(A) + \text{rank}(B)$$